

observed in 33 (55%) patients and grade 2 mandibular complications in 2 (3%) patients. There were no grade 3 or more late complications.

Conclusion: Our 10-year treatment results of IBT for locally advanced tongue cancer showed favorable survival and low occurrence rate in patients with severe late complications.

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Predictors of Outcomes in Iodine-125 Brachytherapy-Treated Uveal Melanomas: A Modern Institutional Study

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Purpose/Objective(s): Plaque brachytherapy is an effective vision- and eye-sparing procedure for uveal melanoma. Few have investigated longitudinal tumor outcomes while using advanced dosimetry and plaque planning software. We hypothesize that patients with large tumors, pre-operative eye complications, and molecular class 2 tumors will have worse outcomes.

Materials/Methods: We constructed an IRB-approved retrospective database from a single academic institution with uveal melanoma patients receiving Iodine-125 brachytherapy, 85 Gy over 72 hours prescribed to the tumor apex, between 2011-2019 with a minimum of 3-month follow-up. Data on patient demographics, tumor characteristics, pre-treatment (pre-Tx) retinal complications, post-plaque treatments (PPTx), local recurrences (LR), distant metastases (DM) and overall survival (OS) were collected. Univariate (UVA) and multivariate (MVA) Cox models for OS and cumulative incidence of LR/DM and death without LR/DM as competing risk were conducted using SAS version 9.4. Hazard ratios (HRs) and corresponding *P*-values were estimated.

Results: 262 patients were identified, with a median follow-up time of 33.5 months (3.02 - 97.3 months). 19 patients (7.3%) had local recurrence, 17 patients (6.5%) had distant metastasis, and 20 patients (7.6%) died. For LR, on UVA pre-Tx ocular melanocytosis (HR = 5.48, *P* = 0.01), Eye Physics plaque (HR = 3.70, *P* = 0.013), and PPTx with transpupillary thermotherapy (HR = 4.65, *P* = 0.002) were associated with increased LR. No association was found between class 2 tumors and LR (HR = 0.51, *P* = 0.297). For DM, on UVA pre-Tx retinal detachment (HR = 7.26, *P* = 0.002), class 2 tumors (HR = 4.35, *P* = 0.005), PPTx with intravitreal triamcinolone (HR = 4.01, *P* = 0.006), and large tumor height (HR = 1.23, *P* < 0.001), tumor area (HR = 1.02, *P* < 0.001), distance to fovea (HR = 1.13, *P* = 0.002), and prescription depth (HR = 1.23, *P* = 0.001) were all associated with increased DM. No MVA was conducted for LR and DM because there were too few events. For any tumor progression (LR or DM), on UVA pre-Tx ocular melanocytosis (HR = 4.26, *P* = 0.005), Eye Physics plaque (HR = 3.41, *P* = 0.002), any PPTx (HR = 4.50, *P* = 0.012), PPTx with intravitreal triamcinolone (HR = 3.30; *P* < 0.01), and large tumor area (HR = 1.01, *P* < 0.001) were associated with progression. On MVA pre-Tx ocular melanocytosis (HR = 5.55, *P* < 0.001), Eye Physics plaque (HR = 2.45, *P* = 0.049), and any PPTx (HR = 4.47, *P* = 0.017) were associated with progression. PPTx with aflibercept was associated with less progression (HR = 0.13, *P* = 0.019).

Conclusion: Ocular melanocytosis and receiving any post-plaque treatment had the greatest impact on tumor progression. Class 2 tumors predicted for distant metastases but not local recurrences. These findings inform physicians of predictors of brachytherapy outcomes and allows for better shared-decision making with pre-operative patients. Future studies must validate these findings using a prospective cohort study.

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Incidence, Treatment and Clinical Outcomes for Lacrimal Gland Cancer

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Purpose/Objective(s): Lacrimal gland cancer is a rare tumor with little data regarding outcomes and optimal management. We performed a large database analysis using the National Cancer Database (NCDB) to elucidate patterns of care and outcomes for this rare malignancy.

Materials/Methods: Patients with lacrimal gland cancer diagnosed between 2004 and 2016 were included in analysis. All histologies were included with the exception of malignant melanoma and lymphoma; squamous cell carcinoma was also excluded as it is rarely found in the lacrimal gland, and likely reflects lacrimal duct or sac cancer. Patients without recorded follow up or with less than 6 months of follow up were excluded. Patients were stratified by sex, age, race, Charlson-Deyo comorbidity score, facility type, and pathologic T stage. Patterns of care including treatment with surgery, chemotherapy and/or radiation therapy were noted. Outcomes were analyzed using Kaplan-Meier analysis, and Cox proportional hazards regression was used to study the effect of surgery and radiation therapy on overall survival.

Results: 460 patients with lacrimal gland cancer were included in the database. Median follow up for all patients was 42.7 months. The most common histology was adenoid cystic carcinoma (185 patients, 40.2%), followed by adenocarcinoma (85, 18.5%). Most patients were of unknown T stage (63.9%). 362 patients (78.7%) underwent surgical resection, 104 (22.6%) received chemotherapy, and 292 (63.5%) underwent radiotherapy. Median overall survival for all patients was 106.8 months (95% CI: 85.3 months - not reached). Variables associated with worse overall survival on univariate analysis included age over 60 (*P* < 0.001), adenocarcinoma histology (*P* < 0.001), and positive surgical margins (*P* = 0.048). Surgical resection was associated with a significant improvement in overall survival at 4 years, from 59.9% (95% CI: 49.9-71.9) to 77.4% (95% CI: 72.7-82.4). Surgical resection was also associated with an improvement in overall survival on univariate (HR 0.487, 95% CI 0.343-0.691, *P* < 0.001) and multivariate regression (HR 0.468, 95% CI 0.313-0.698, *P* < 0.001). Receiving radiotherapy did not appear to significantly affect overall survival (HR 1.064, 95% CI 0.758-1.495, *P* = 0.719), even for a subset of patients who underwent surgical resection with T4 cancers or positive margins (HR 1.219, 95% CI 0.651-2.286, *P* = 0.536).

Conclusion: The ideal management for lacrimal gland cancer is still unknown due to the rarity of this tumor. Retrospective study of a large national database suggests that patients who undergo surgical resection may have better outcomes than those who do not, although a benefit for radiotherapy is not apparent. Further data collection is required for this unusual tumor.

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Long Term Clinical Outcomes With Dedicated Proton Ocular Beam and Helium Ion Radiation for Uveal Melanoma Patients Aged 45 and Younger

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